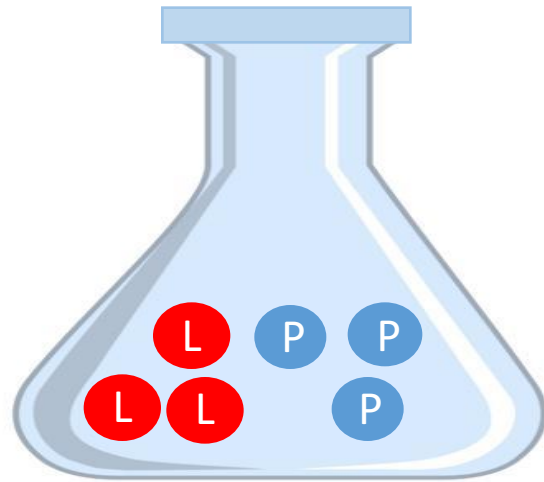


**Summary from key concepts
&
Nutrients**

Dr. P. Satpati,
BSBE, IIT Guwahati

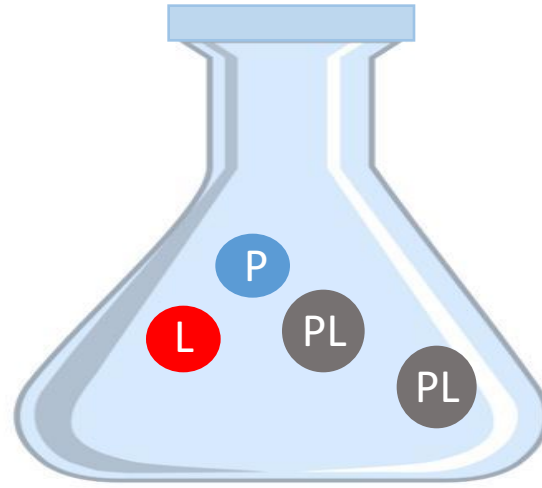
Thermodynamics and kinetics

P= Protein
L= Ligand
PL= complex



Time = 0

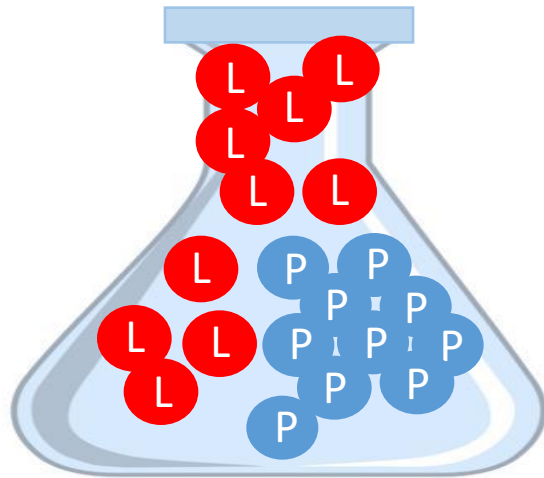
Time →



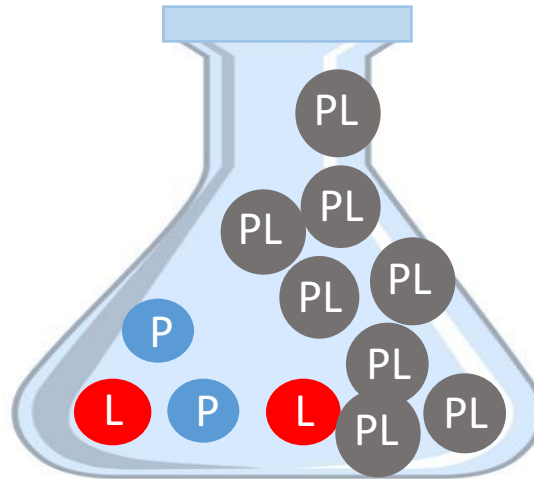
Equilibrium (at Time = t)

$$K_{eq} = \frac{[PL]_{eq}}{[P]_{eq}[L]_{eq}} = \frac{2}{1 \times 1} = 2$$

At equilibrium the ratio of concentration = constant



Time →



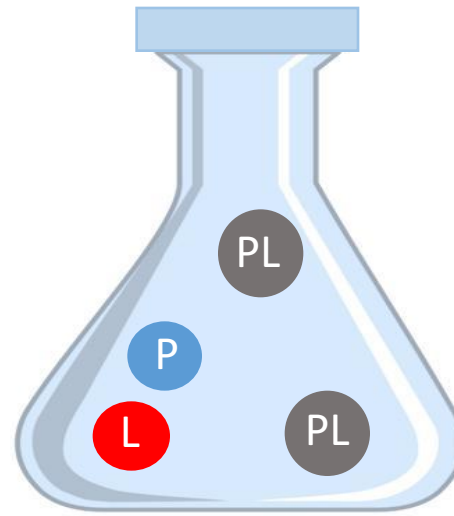
$$K_{eq} = \frac{[PL]_{eq2}}{[P]_{eq2}[L]_{eq2}} = \frac{8}{2 \times 2} = 2$$

Equilibrium constant is independent of the initial concentrations of the reactant and product.

What influence the value of K_{eq} ?

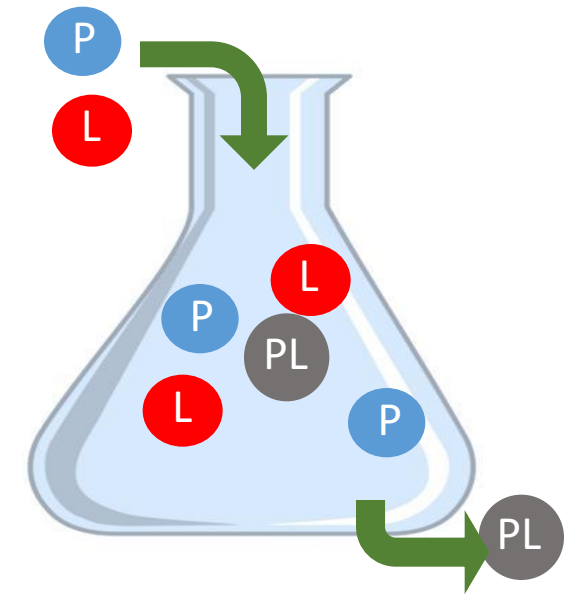
- Temperature
- Nature of solvent
- Salt (buffer-condition)
- Pressure

$$K_{eq}(T)$$



Equilibrium

$$K_{eq} = 2$$



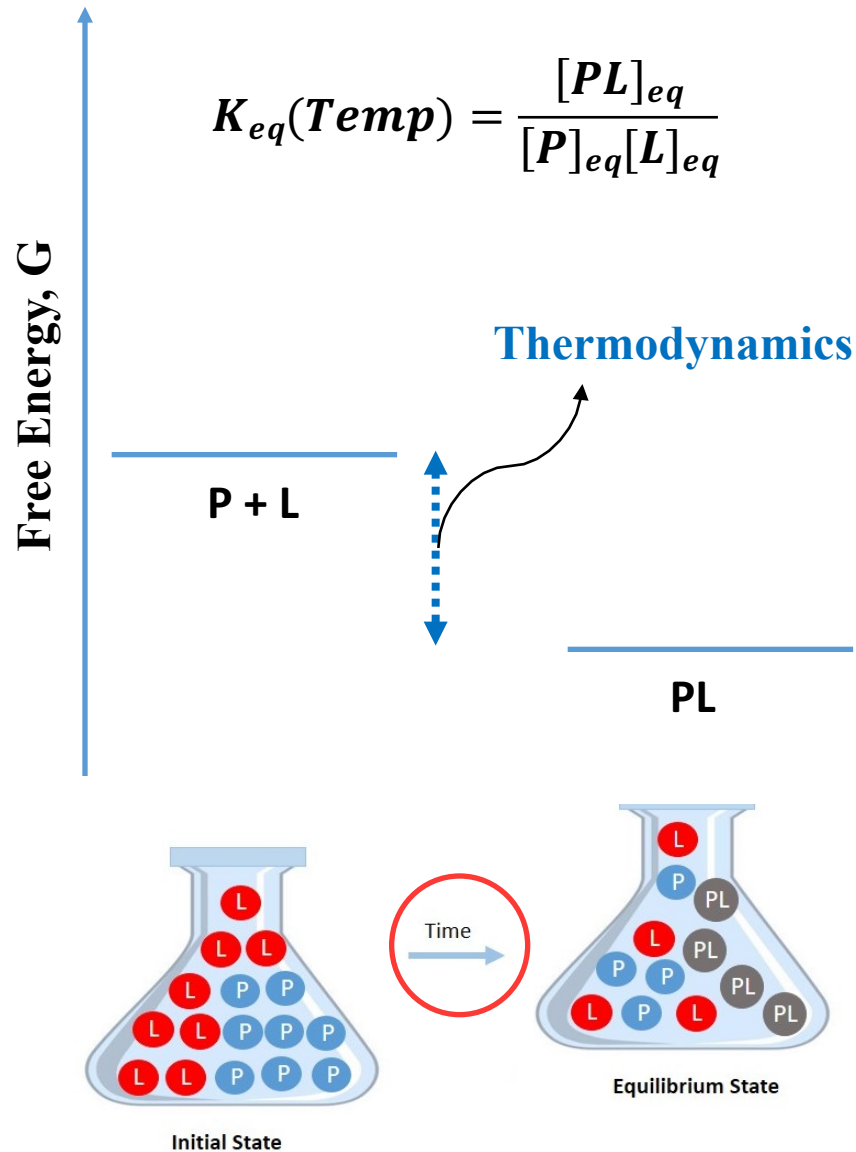
Steady State

$$K' = \frac{1}{4}$$

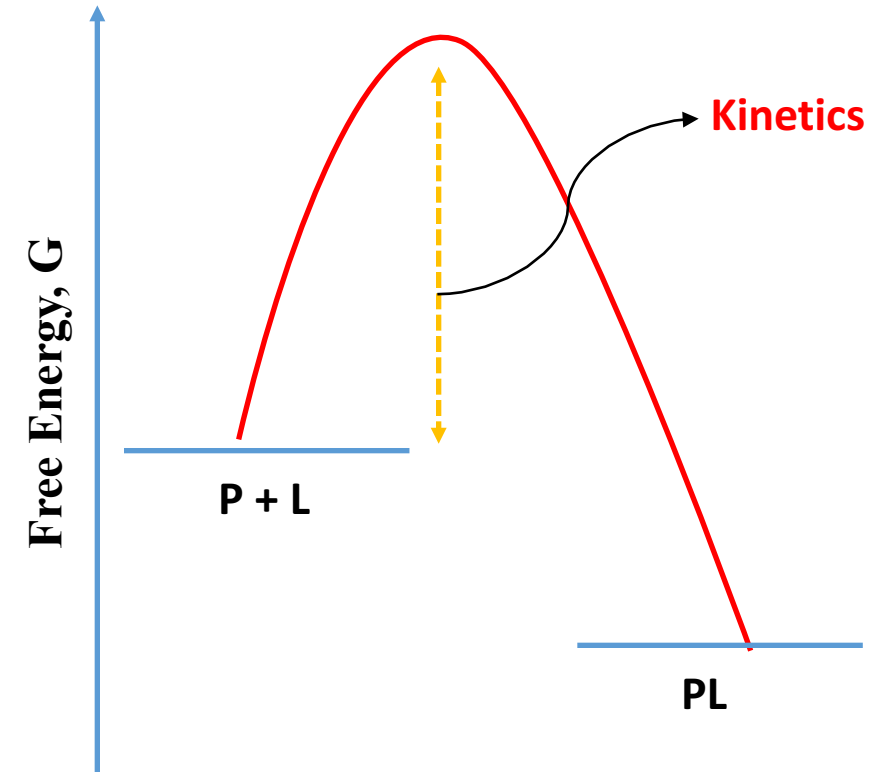
> **Equilibrium $\neq$ Steady State**

> **Steady State is maintained by a flow (P, L in; PL out)**

- What defines the K_{eq} / concentration ratio ?

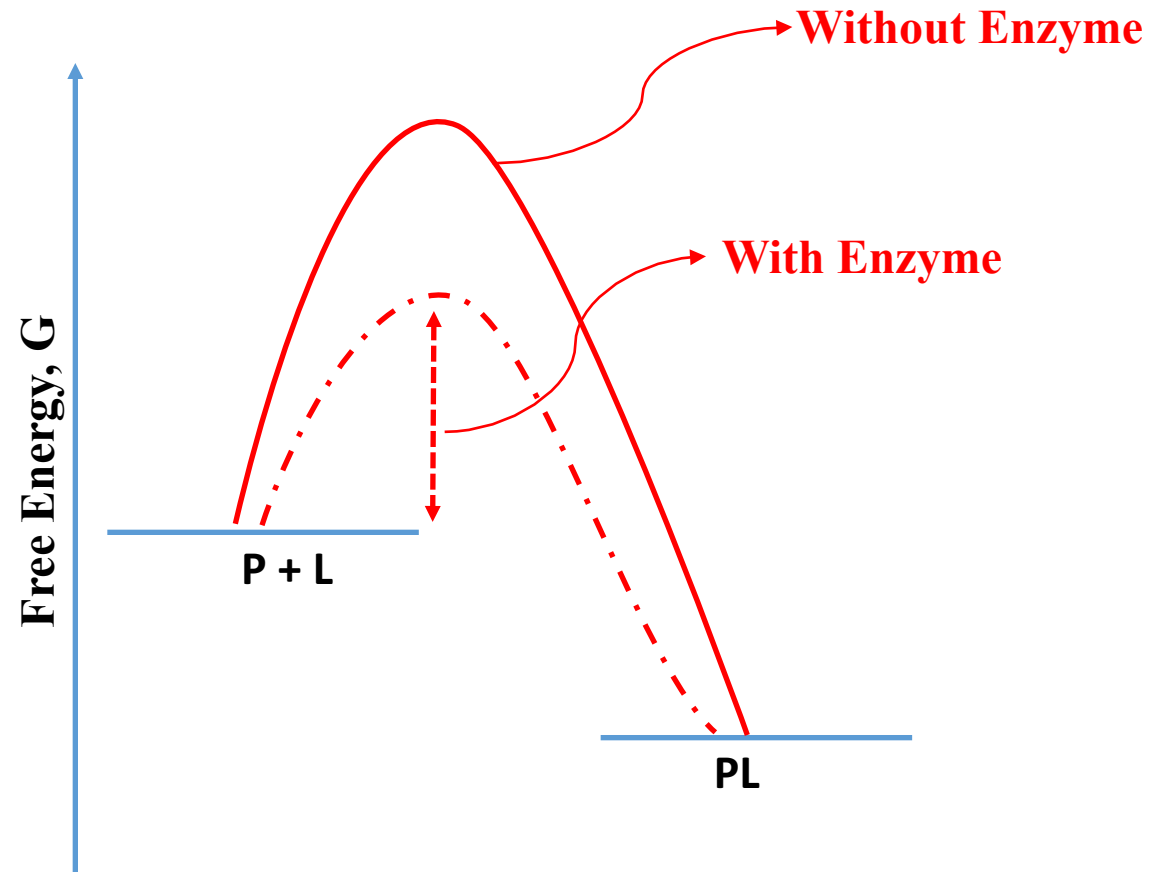


- What defines the time ?



- Path connecting reactants and products (activation barrier). **KINETICS**
- Large barrier = More Time requires to reach equilibrium

■ What speed up the reactions in biology ?



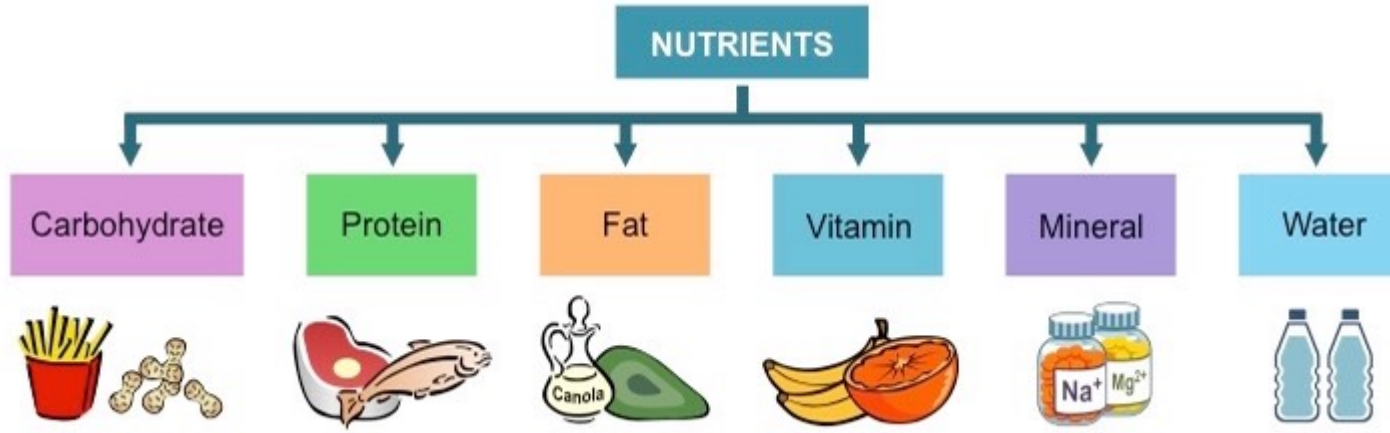
- Enzymes reduces the activation barrier, and speed up the reaction.

Nutrients

Nutrition ?

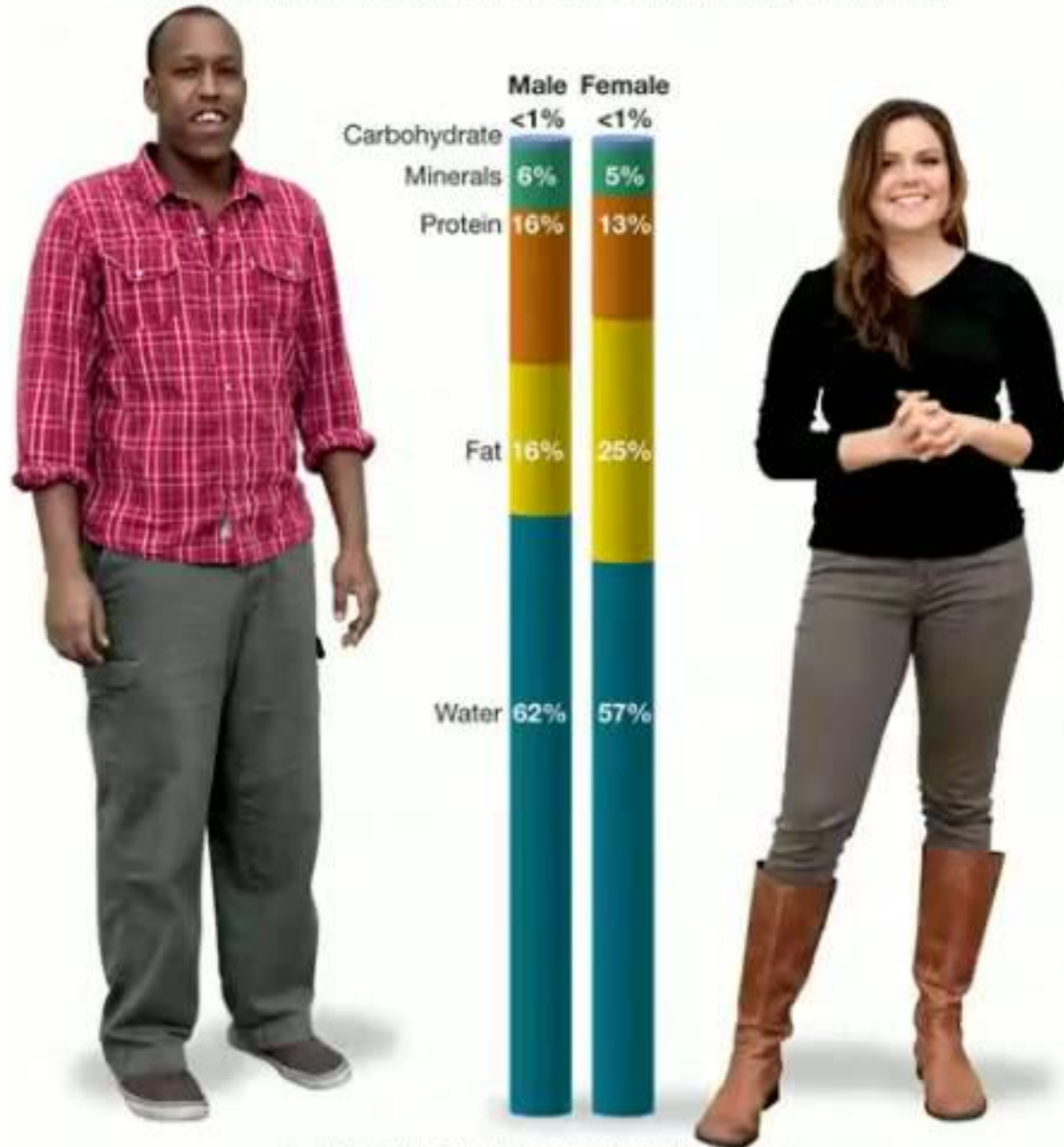
- Nutrition is the science that studies the process by which living organism **acquire** all the things that are necessary For them to **live** and **grow**
- Nutrition focuses on the role of **nutrients** (**Molecules in food needed to make energy, grow, develop, reproduce**)
 - Nutrition consist of diet (what you eat)
 - Metabolism (What happens to the food after it enters to your body)
 - Catabolism: Catabolic reactions **break down** bigger molecules and **release energy**
 - Anabolism: Anabolic reactions **construct** things and **consume energy**

The six classes of nutrients



Both Human body and food are comprised of nutrients

Nutrient	Primary functions
Water	Dissolves and carries nutrients, removes waste, and regulates body temperature
Protein	Builds new tissues, antibodies, enzymes, hormones, and other compounds
Carbohydrate	Provides energy
Fat	Provides long-term energy, insulation, and protection
Vitamins	Facilitate use of other nutrients; involved in regulating growth and manufacturing hormones
Minerals	Help build bones and teeth; aid in muscle function and nervous system activity



- Men and women have similar amounts of vitamins, minerals, and carbohydrates.
- Women have less water and protein and more fat.
- 50 to 70% of total body weight is water.

ORGANIC Nutrients : Contains Carbon
INORGANIC Nutrients: NO Carbon

		Carbon	Hydrogen	Oxygen	Nitrogen	Single elements
Organic	Carbohydrates	√	√	√		
	Lipids	√	√	√		
	Proteins	√	√	√	√	
	Vitamins	√	√	√	√ (some)	
Inorganic	Minerals					√
	Water		√	√		

Macronutrients- Required in Large amount

Micronutrients - Required in small amount (Vitamins, Minerals)

Essentiality

Essential nutrients: Cannot be synthesized in the body (Must be included through diet). In the case of humans, Apart from water 9 amino acids, 2 two fatty acids, 13 vitamins, and 15 minerals are considered essential nutrients.

Non-essential nutrients: Can be made by the body or obtained from sources other than foods and beverages.

e.g, - **biotin** that is produced by gastrointestinal bacteria,

- **cholesterol** that is produced by the liver,
- **vitamin K** that is produced by intestinal bacteria,
- **vitamin D** that is produced by sunlight.
- **Dietary fiber** is not absorbed in the human digestive tract, but is important in maintaining the bulk of a bowel movement to avoid constipation.
- A subset of dietary fiber, **soluble fiber**, can be metabolized by bacteria residing in the large intestine.

<https://en.wikipedia.org/wiki/Nutrient>

<https://www.sciencedirect.com/topics/biochemistry-genetics-and-molecular-biology/essential-nutrient>

Energy-yielding nutrients



Carbohydrates	√
Lipids	√
Proteins	√
Vitamins	X
Minerals	X
Water	X

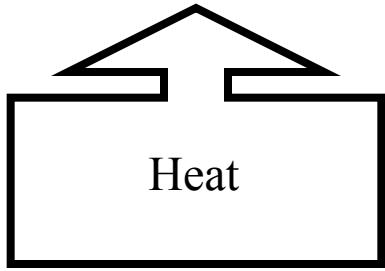
Energy Currency  = Calorie / Kilocalorie


Calories per gram

Carbohydrates	√	4
Lipids	√	9
Proteins	√	4
Vitamins	X	0
Minerals	X	0
Water	X	0

What is Calorie (Cal) or Kilocalorie (Kcal) ?

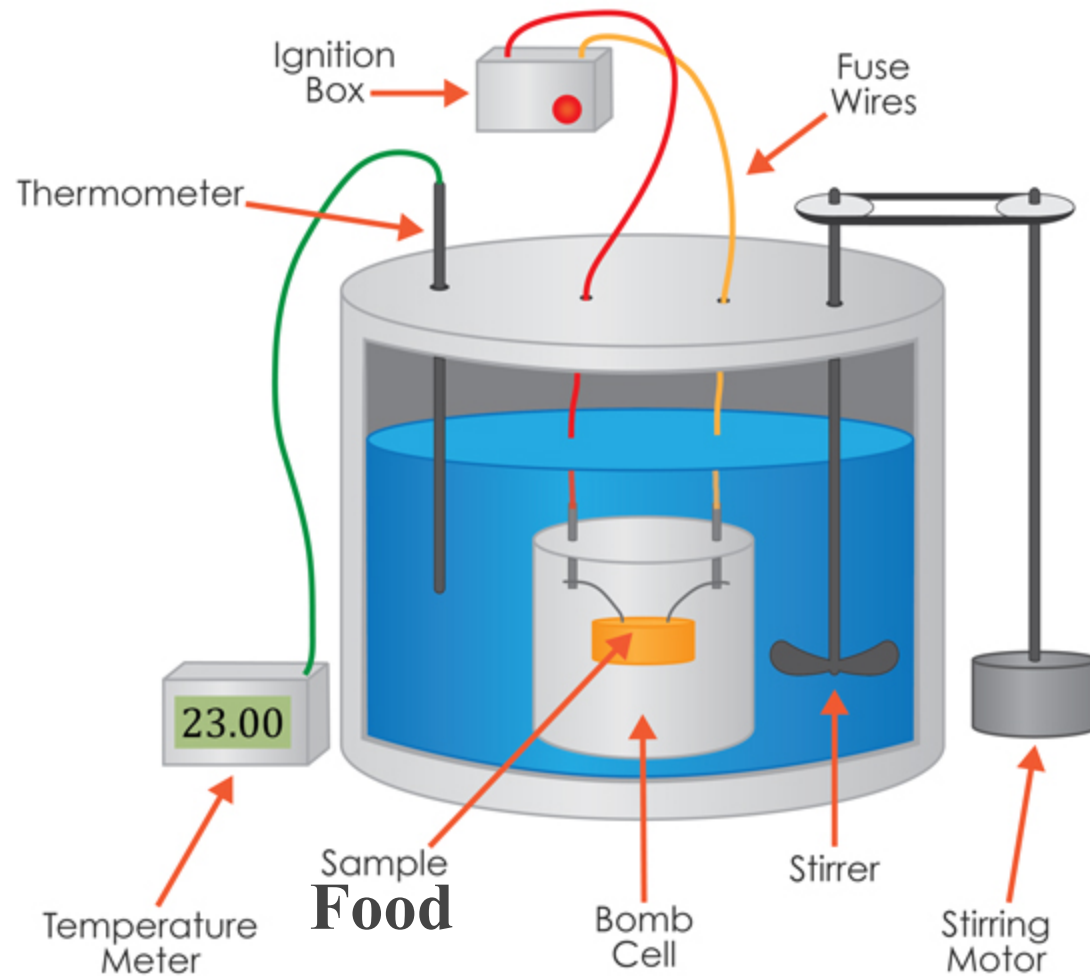
1 Cal = Energy required to raise the temperature of 1 gm of water by 1 degree Celsius



1 Kcal = Energy required to raise the temperature of 1000gm of water by 1 degree Celsius

How to determine energy content (kcal) of food?

Bomb Calorimeter



$$Q \propto \Delta T$$

$$Q = C M \Delta T$$

Q = Heat [Cal]

C = Specific heat of water [1 cal/(gm $^{\circ}$ C)]

M = Mass of liquid [gm]

ΔT = Change in temperature [$^{\circ}$ C]

<https://www.youtube.com/watch?v=cw7q433ynYg>



Ingredients: Potatoes, Vegetable Oil (Sunflower, Corn and/or Canola Oil), and Salt.

Nutrition Facts

Serving Size 1 oz (28g/About 15 chips)

Amount Per Serving

Calories 160 Calories from Fat 90

% Daily Value*

Total Fat 10g **16%**

Saturated Fat 1.5g **8%**

Trans Fat 0g

Cholesterol 0mg **0%**

Sodium 170mg **7%**

Potassium 350mg **10%**

Total Carbohydrate 15g **5%**

Dietary Fiber 1g **5%**

Sugars less than 1g

Protein 2g

Vitamin A 0% • Vitamin C 10%

Calcium 0% • Iron 2%

Vitamin E 6% • Thiamin 4%

Niacin 6% • Vitamin B₆ 10%

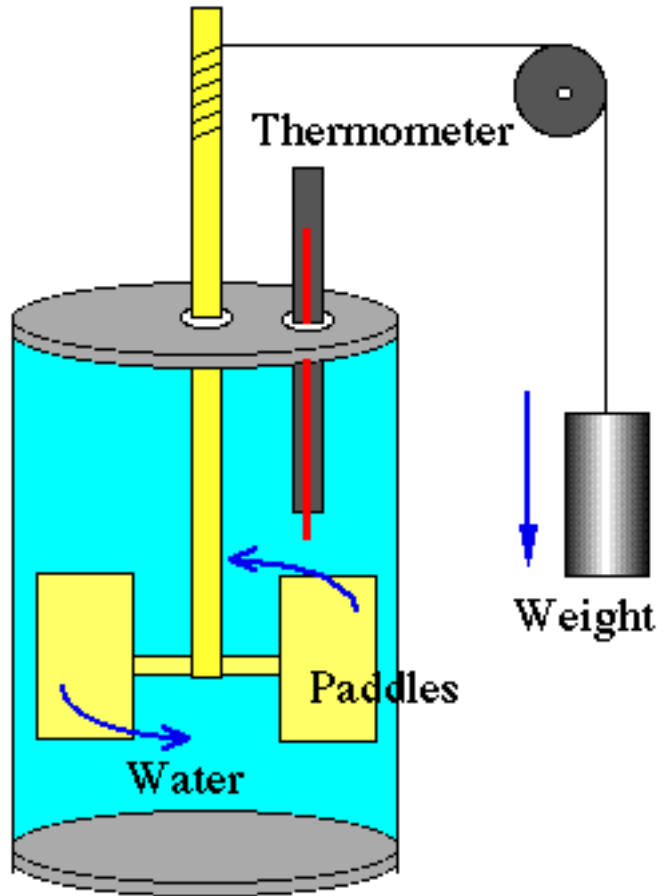
Magnesium 4% • Zinc 2%

* Percent Daily Values are based on a 2,000 calorie diet. Your daily values may be higher or lower depending on your calorie needs:

		Calories:	2,000	2,500
Total Fat	Less than		65g	80g
Sat Fat	Less than		20g	25g
Cholesterol	Less than		300mg	300mg
Sodium	Less than		2,400mg	2,400mg
Potassium			3,500mg	3,500mg
Total Carbohydrate			300g	375g
Dietary Fiber			25g	30g

Calories per gram:

Fat 9 • Carbohydrate 4 • Protein 4

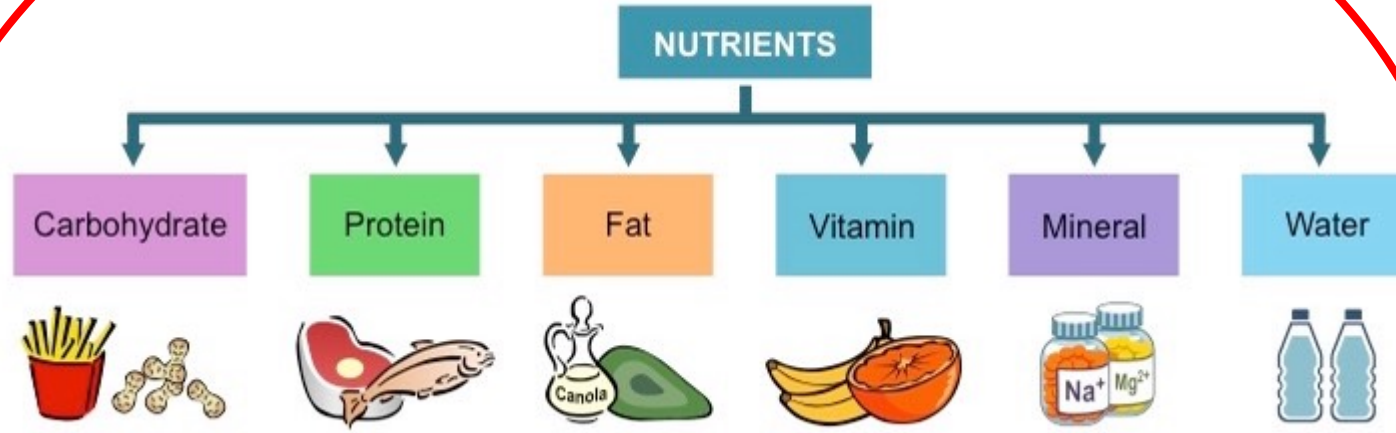


Joule's Experiment

Heat = Mechanical Energy

1 cal = 4.2 Joule

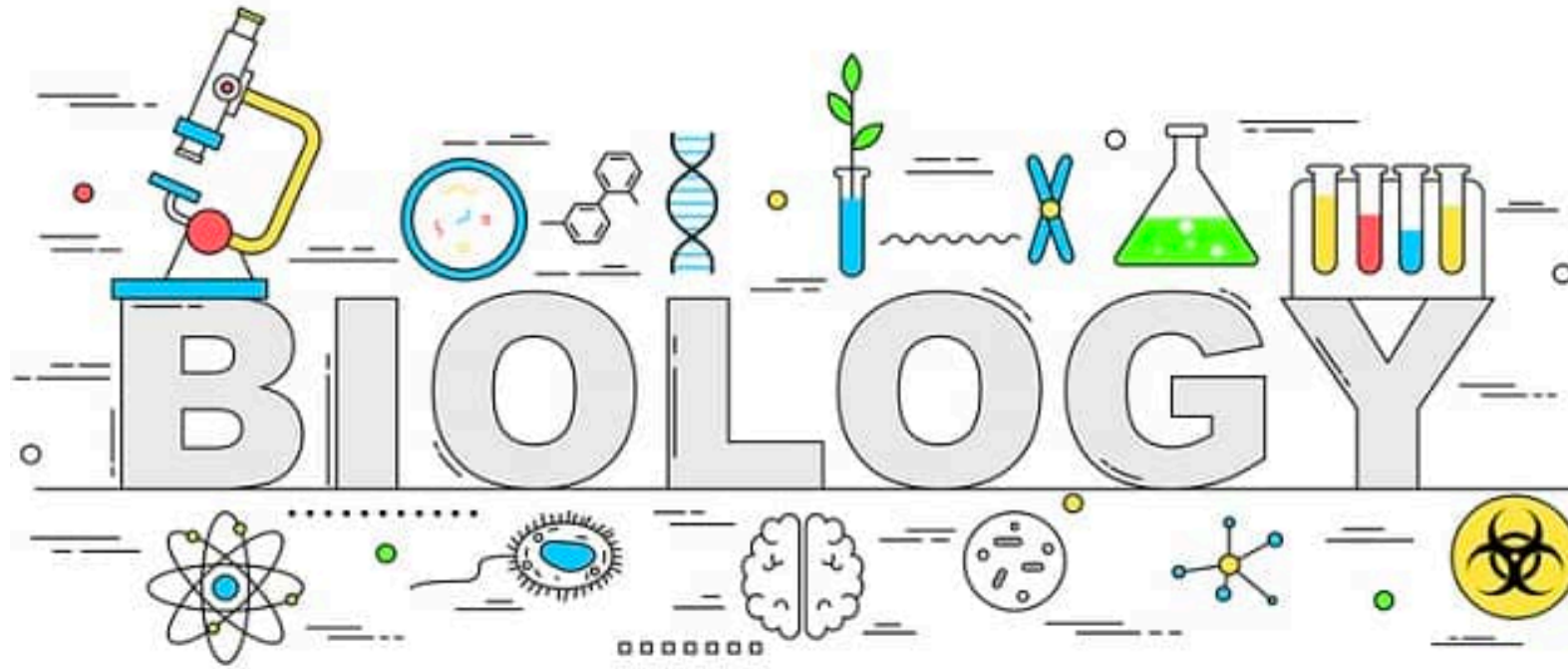
The six classes of nutrients



Both Human body and food are comprised of nutrients

Nutrient	Primary functions
Water	Dissolves and carries nutrients, removes waste, and regulates body temperature
Protein	Builds new tissues, antibodies, enzymes, hormones, and other compounds
Carbohydrate	Provides energy
Fat	Provides long-term energy, insulation, and protection
Vitamins	Facilitate use of other nutrients; involved in regulating growth and manufacturing hormones
Minerals	Help build bones and teeth; aid in muscle function and nervous system activity

Let's discuss one by one



Next: Carbohydrates